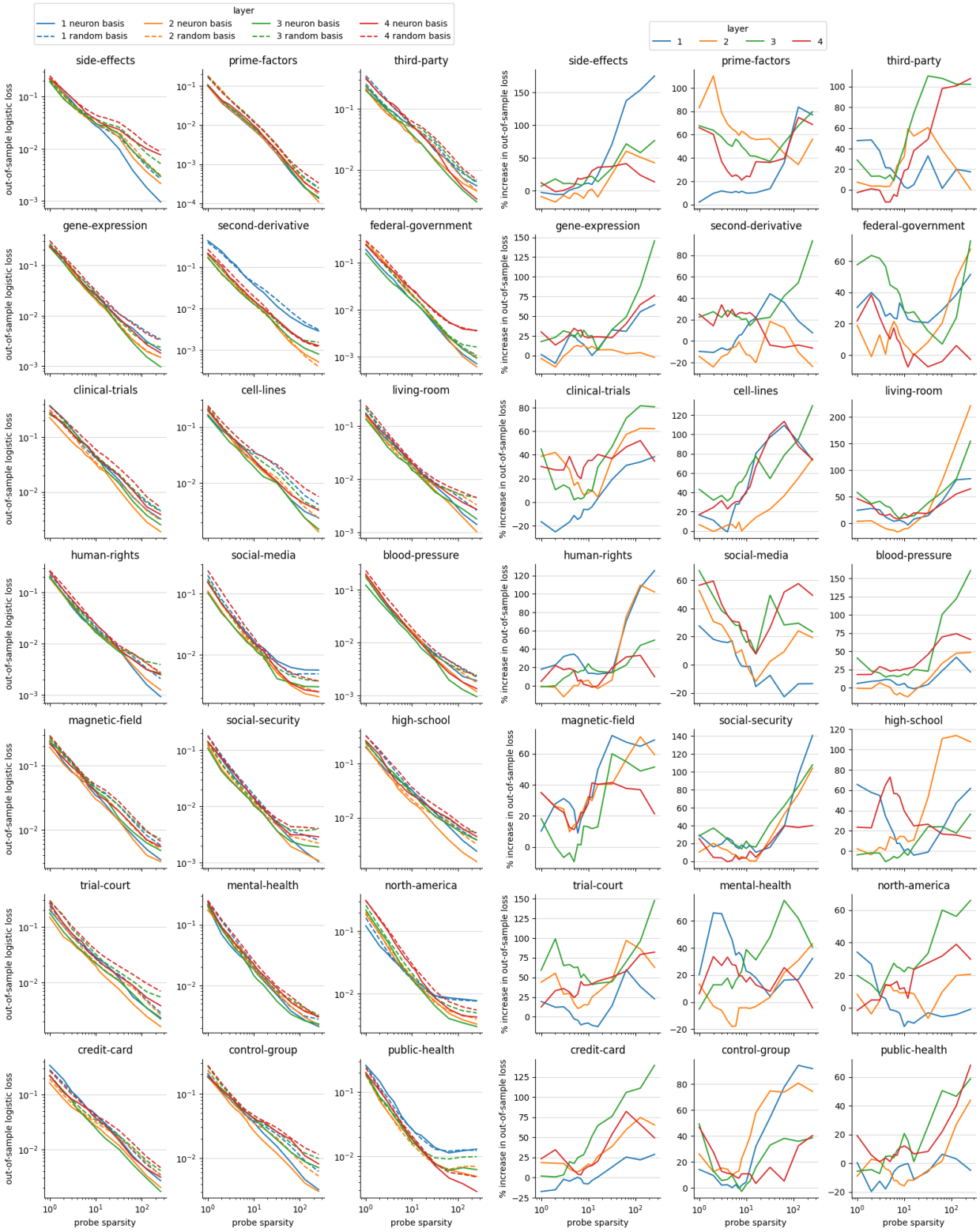

(a) Logistic loss versus probe sparsity

(b) Logistic loss increase versus probe sparsity

Figure 14: Logistic loss for classifying the occurrence of a particular compound word in layers 1-4 of Pythia-1B (in both the neuron basis and the average of random bases) undergo power law scaling until hitting a break point (left). In general, probes in the neuron basis achieve lower overall loss, further suggesting the neuron basis is meaningful (right).

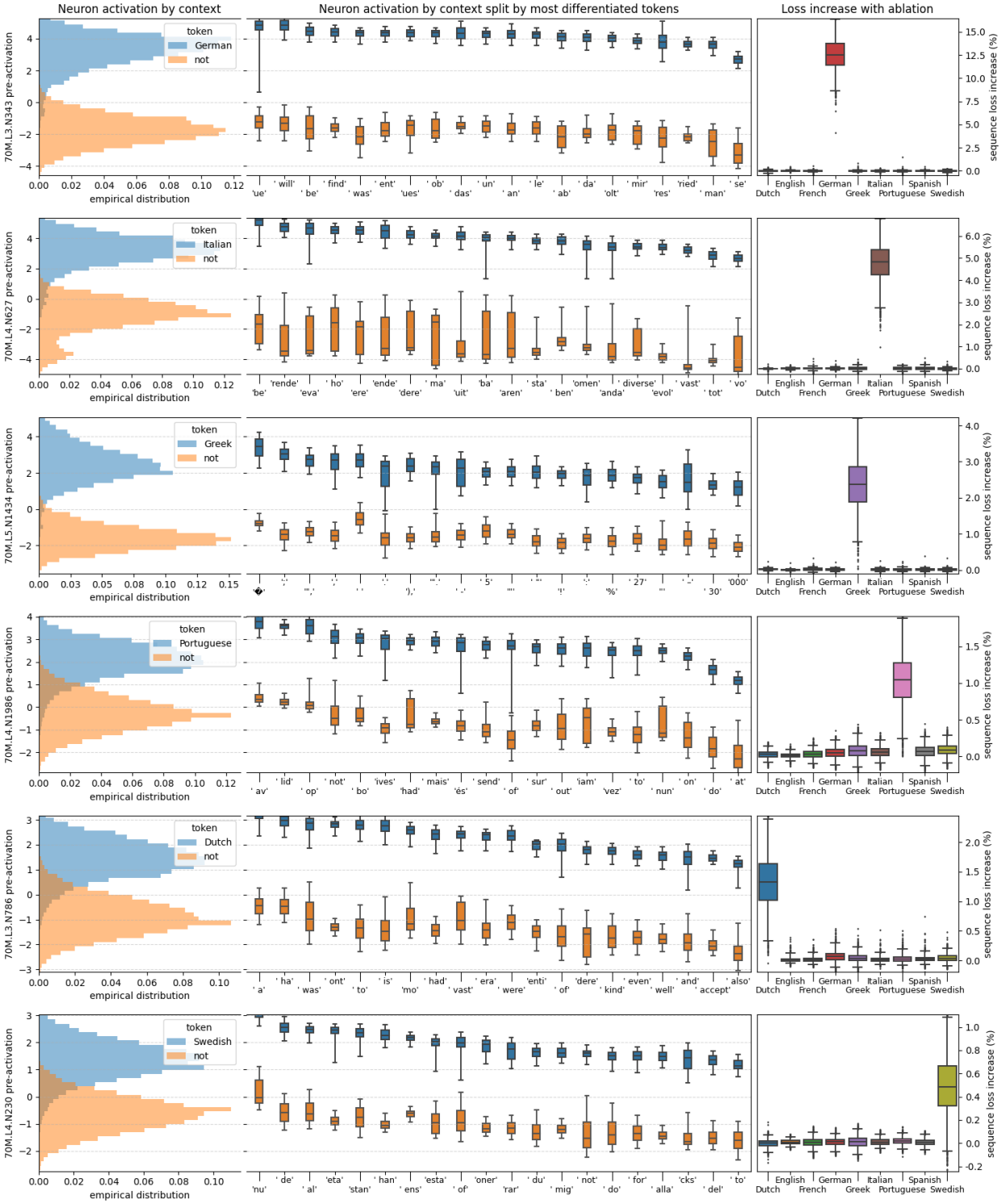

Figure 15: Monosemantic language context neurons in Pythia-70M.

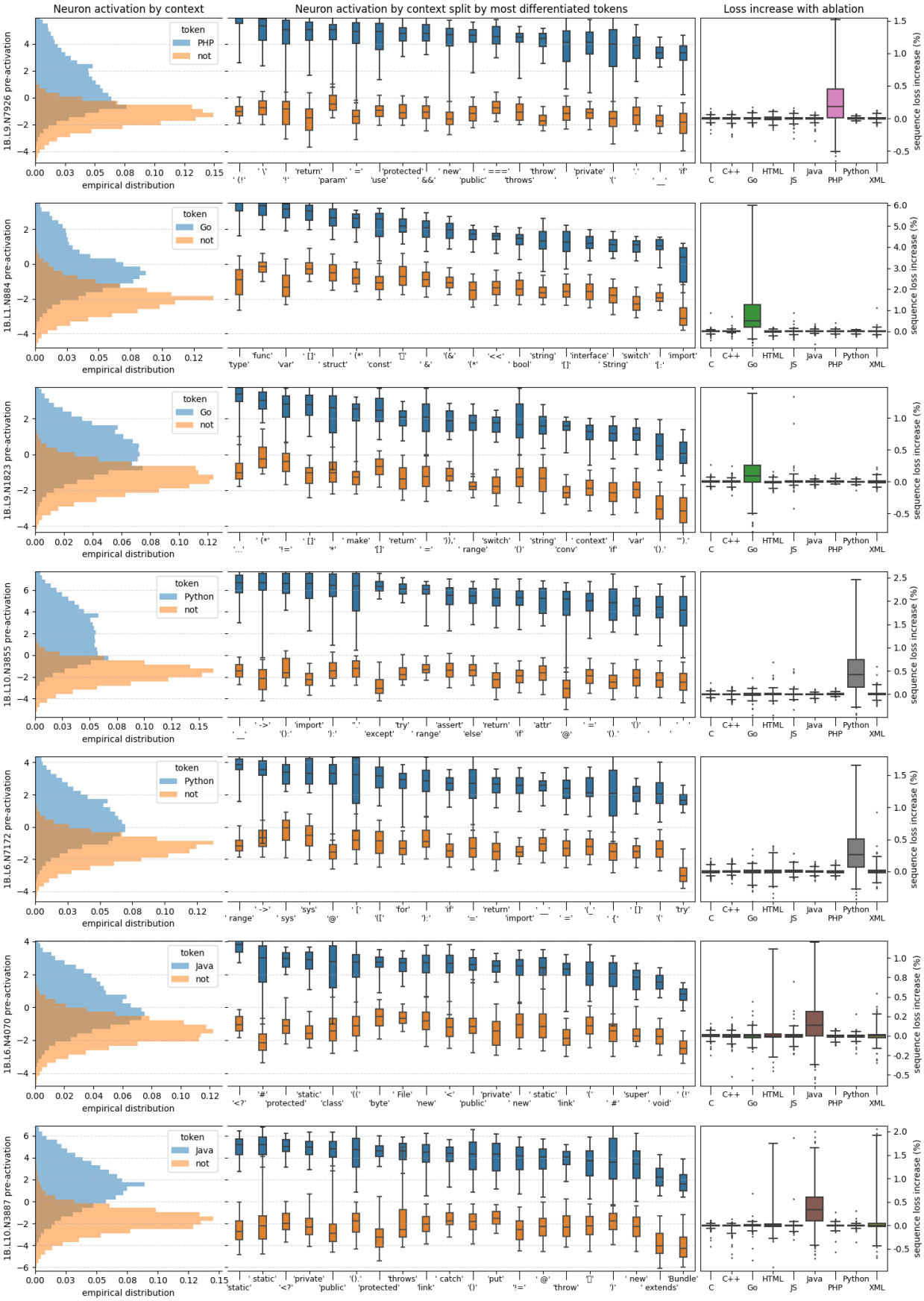

Figure 16: Monosemantic code context neurons in Pythia-1B.